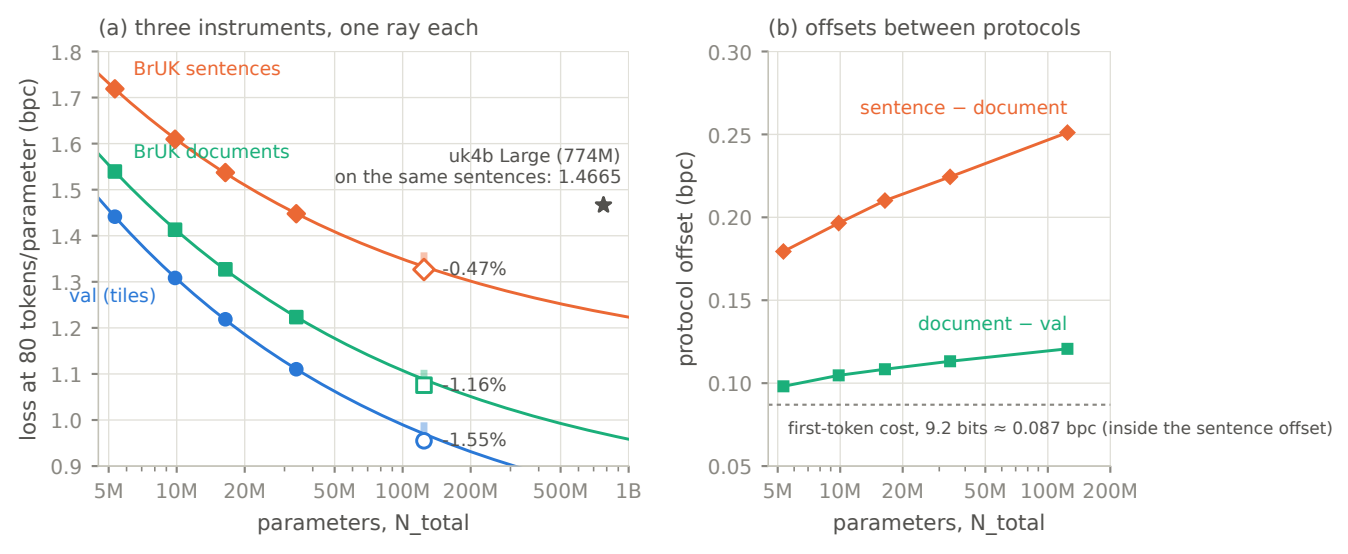
(a) three instruments, one ray each
BrUK sentences
BrUK documents
val (tiles)
uk4b Large (774M)
on the same sentences: 1.4665
-0.47%
-1.16%
-1.55%
loss at 80 tokens/parameter (bpc)
parameters, N_total
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
5M
10M
20M
50M
100M
200M
500M
1B
(b) offsets between protocols
sentence − document
document − val
first-token cost, 9.2 bits ≈ 0.087 bpc (inside the sentence offset)
protocol offset (bpc)
parameters, N_total
0.30
0.25
0.20
0.15
0.10
0.05
5M
10M
20M
50M
100M
200M